ZETDC DocTel LLM

Functional Requirements Specification (FRS)

DocTel Mobile – Multi-Open-Source AI Document Intelligence Platform

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Organisation: Zimbabwe Electricity Transmission and Distribution Company (ZETDC)
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1. Introduction

1.1 Purpose

This document defines the functional requirements for the mobile version of the DocTel Application, an AI-powered document intelligence platform designed for Android and iOS devices.

The mobile platform shall provide full feature parity with the web-based DocTel system while extending support for multiple open-source AI models, offline-aware processing, mobile-native workflows, and secure enterprise collaboration.

This specification serves as the authoritative reference for:

- Mobile application developers
- Backend/API developers
- QA and testing teams
- Infrastructure engineers
- Security auditors
- ZETDC business stakeholders

1.2 Scope

The DocTel Mobile Application enables authorised ZETDC personnel to:

- Upload and manage documents
- Perform AI-powered document analysis
- Chat with documents using Retrieval-Augmented Generation (RAG)
- Access repository-wide and organisation-wide intelligence
- Manage AI models and integrations
- Generate reports, charts, summaries, and policy drafts
- Perform collaborative analysis workflows
- Use multiple open-source AI engines locally and remotely
- Access Training Room functionality
- Operate securely in both online and restricted/offline environments

The mobile solution shall support:

- Android phones and tablets
- iPhones and iPads
- React Native (Expo)
- REST API integration
- Local caching and synchronisation
- Push notifications
- Background processing
- Secure token-based authentication

1.3 Supported Open-Source Technologies

The mobile application and backend shall support integration with multiple open-source technologies including but not limited to:

Category	Supported Technologies
LLM Runtime	Ollama, llama.cpp, vLLM, LocalAI
Models	LLaMA, Qwen, DeepSeek, Mistral, Gemma, Phi
Embeddings	Nomic Embed, BGE, InstructorXL
Vector Database	ChromaDB, Qdrant, FAISS, Weaviate
OCR Engines	Tesseract OCR, PaddleOCR
Speech-to-Text	Whisper
Text-to-Speech	Piper TTS
Mobile Storage	SQLite, RealmDB
Authentication	LDAP/Active Directory, OAuth2, JWT

The system architecture shall allow future extensibility for additional open-source integrations.

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2. Mobile System Architecture

FR-MARCH-01 – Cross-Platform Framework

The mobile application shall be developed using React Native with Expo support for Android and iOS.

FR-MARCH-02 – Backend Connectivity

The mobile app shall communicate with the FastAPI backend using secure REST APIs and WebSocket/SSE streaming.

FR-MARCH-03 – Offline Cache

The mobile application shall locally cache:

- User sessions
- Recent documents
- Chat history
- AI outputs
- Repository metadata
- Settings

Cached data shall synchronise automatically when connectivity is restored.

FR-MARCH-04 – Modular AI Provider Support

The backend shall expose pluggable AI provider interfaces allowing switching between:

- Ollama
- llama.cpp
- LocalAI
- vLLM
- Gemini
- OpenAI-compatible endpoints
- Future custom providers

without requiring mobile application changes.

FR-MARCH-05 – API Compatibility

All APIs available to the web application shall also be consumable by the mobile application.

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3. User Roles & Access Control

FR-MRBAC-01 – Role Definitions

The mobile application shall support the following user roles:

- Admin
- Analyst
- Viewer

FR-MRBAC-02 – Mobile Role Enforcement

All mobile actions shall enforce server-side RBAC validation.

FR-MRBAC-03 – Repository-Level Permissions

Users shall have repository-scoped permissions inherited from backend configuration.

FR-MRBAC-04 – Mobile Session Security

Role and permission validation shall occur during token refresh and session restoration.

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4. Authentication & Security

FR-MAUTH-01 – Email OTP Login

Users shall authenticate using OTP verification sent to approved organisational email addresses.

FR-MAUTH-02 – EC Number Login

The mobile application shall support EC Number and password login through LDAP/Active Directory.

FR-MAUTH-03 – Biometric Authentication

The mobile app shall support:

- Fingerprint authentication
- Face ID
- Android Biometrics

for session unlock.

FR-MAUTH-04 – Secure Token Storage

JWT access and refresh tokens shall be securely stored using encrypted device storage.

FR-MAUTH-05 – Auto Session Renewal

The application shall automatically refresh valid sessions in the background.

FR-MAUTH-06 – Session Timeout

Inactive sessions shall automatically expire after the configured inactivity period.

FR-MAUTH-07 – Device Registration

The backend may optionally track registered mobile devices including:

- Device ID

- Platform
- Last login time
- App version
- Push notification token

FR-MAUTH-08 – Remote Logout

Admin users may remotely revoke active mobile sessions.

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5. Mobile Dashboard

FR-MDASH-01 – Dashboard Overview

The mobile home dashboard shall display:

- Recent documents
- Recent chat sessions
- Repository summaries
- Processing status
- AI model status
- Notifications
- Quick actions

FR-MDASH-02 – Personalised Widgets

Users may customise dashboard widgets and layouts.

FR-MDASH-03 – Global Search

The dashboard shall include global search across:

- Documents
- Repositories
- Chats
- Outputs
- Reports
- Users (admin only)

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6. Document Management

FR-MDOC-01 – Mobile File Upload

Users shall upload files from:

- Device storage

- Camera capture
- Photo gallery
- Cloud storage providers
- Email attachments

FR-MDOC-02 – Supported File Types

The mobile application shall support:

- PDF
- DOCX
- XLSX
- PPTX
- TXT
- CSV
- PNG
- JPEG
- WEBP
- GIF
- BMP
- Audio files
- ZIP archives

FR-MDOC-03 – Multi-File Upload

Users shall upload multiple files simultaneously.

FR-MDOC-04 – Metadata Assignment

Users may assign:

- Repository
- Tags
- Document type
- Document date
- Classification level
- Priority

before upload.

FR-MDOC-05 – Camera Scan Mode

The mobile app shall provide a built-in document scanning mode supporting:

- Auto edge detection
- Perspective correction
- OCR pre-processing
- Multi-page scan
- PDF export

FR-MDOC-06 – Drag-and-Drop Support

Tablet devices shall support drag-and-drop uploads where supported by the OS.

FR-MDOC-07 – Document Preview

Users shall preview documents directly within the mobile app.

FR-MDOC-08 – Document Annotation

Users may:

- Highlight text
- Add comments
- Draw annotations
- Save markups

on supported document types.

FR-MDOC-09 – Offline Upload Queue

Documents uploaded while offline shall be queued and automatically synchronised later.

FR-MDOC-10 – SHA-256 Deduplication

Uploaded documents shall be deduplicated using SHA-256 hashing.

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7. Document Ingestion Pipeline

FR-MING-01 – Ingestion Monitoring

Users shall monitor ingestion progress in real time.

FR-MING-02 – Mobile Notifications

Push notifications shall be generated for:

- Upload completion
- Ingestion success
- Ingestion failure
- Analysis completion

FR-MING-03 – OCR Processing

The system shall support OCR using:

- Gemini Vision
- Tesseract OCR
- PaddleOCR

based on availability and policy configuration.

FR-MING-04 – Background Processing

Uploads and ingestion tasks shall continue when the mobile application is minimised.

FR-MING-05 – Retry Support

Users may retry failed ingestion jobs.

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8. AI Chat & RAG Copilot

FR-MCHAT-01 – Document Chat

Users shall ask natural-language questions about a selected document.

FR-MCHAT-02 – Repository Chat

Users shall perform repository-wide semantic chat.

FR-MCHAT-03 – Organisation Chat

Users shall perform organisation-wide chat across authorised repositories.

FR-MCHAT-04 – Voice Input

The mobile app shall support voice-to-text chat prompts using:

- Whisper
- Native mobile speech APIs

FR-MCHAT-05 – Streaming Responses

AI responses shall stream incrementally in real time.

FR-MCHAT-06 – Citation Display

Responses shall display inline source citations.

FR-MCHAT-07 – Suggested Prompts

The chat interface shall show contextual prompt suggestions.

FR-MCHAT-08 – Multi-Model Selection

Users may switch between available AI models per session.

FR-MCHAT-09 – Session Persistence

Chat history shall persist across devices and sessions.

FR-MCHAT-10 – Chat Export

Users may export chat conversations as:

- PDF
- DOCX
- Markdown
- JSON

FR-MCHAT-11 – Voice Responses

The system may optionally read responses aloud using TTS engines.

FR-MCHAT-12 – Multi-Language Support

The chat system shall support multilingual interactions.

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9. AI Model Management

FR-MMODEL-01 – Open-Source Model Support

The system shall support multiple open-source LLMs including:

- LLaMA
- Qwen
- DeepSeek
- Gemma
- Mistral
- Phi

FR-MMODEL-02 – Dynamic Model Routing

The backend shall dynamically route requests based on:

- RAM availability
- GPU availability
- Query complexity
- Offline policy
- Confidence scoring

FR-MMODEL-03 – Local AI Priority

When offline_only=true the system shall use only local/open-source models.

FR-MMODEL-04 – Mobile Model Selector

The mobile interface shall expose a model selector per session.

FR-MMODEL-05 – Model Health Monitoring

The system shall display model availability and health status.

FR-MMODEL-06 – Downloadable Mobile Models

Future releases may support on-device compact AI models.

FR-MMODEL-07 – AI Provider Fallback

The system shall support fallback routing between providers.

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10. Analysis Tools

FR-MTOOL-01 – Multi-Document Summarisation

Users may summarise multiple documents simultaneously.

FR-MTOOL-02 – Data Extraction

Users may define extraction schemas.

FR-MTOOL-03 – Document Comparison

Users may compare two or more documents.

FR-MTOOL-04 – Classification Engine

The system shall classify documents using AI-defined rules.

FR-MTOOL-05 – Sentiment Analysis

The system shall identify sentiment and urgency.

FR-MTOOL-06 – Entity Recognition

The system shall extract:

- Names
- Organisations
- Dates
- Locations
- Financial values
- Policy references

FR-MTOOL-07 – Policy Generation

The AI assistant shall generate structured policy drafts.

FR-MTOOL-08 – Action Item Extraction

The system shall identify action items and decisions.

FR-MTOOL-09 – Batch Operations

Users may apply analysis operations to multiple documents.

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11. Diagram & Chart Generation

FR-MDIAG-01 – Mermaid Diagram Generation

Users may generate Mermaid flowcharts from prompts or documents.

FR-MDIAG-02 – Chart Builder

Users may upload CSV/XLSX files and generate charts.

FR-MDIAG-03 – Interactive Charts

Charts shall support:

- Zoom
- Filtering
- Export
- Full-screen viewing

FR-MDIAG-04 – Image Export

Generated diagrams and charts may be exported as PNG or PDF.

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12. Collaboration Features

FR-MCOLLAB-01 – Shared Repositories

Users may collaborate within repositories.

FR-MCOLLAB-02 – Shared Chat Sessions

Users may share AI conversations.

FR-MCOLLAB-03 – Activity Feed

The mobile app shall display repository activity.

FR-MCOLLAB-04 – Mentions & Notifications

Users may mention team members in comments.

FR-MCOLLAB-05 – Push Notifications

The app shall support push notifications for:

- Shared documents
- Mentions
- Completed analysis
- Training events
- System alerts

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13. Outputs & Reporting

FR-MOUT-01 – Output Repository

Generated outputs shall be stored and retrievable.

FR-MOUT-02 – Export Formats

Users shall export outputs as:

- PDF
- DOCX
- JSON
- Markdown
- CSV

FR-MOUT-03 – Scheduled Reports

The system may support scheduled report generation.

FR-MOUT-04 – Mobile Report Viewer

Reports shall be viewable within the mobile app.

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14. Training Room

FR-MTRAIN-01 – Mobile Training Dashboard

Admin users shall access training dashboards on mobile.

FR-MTRAIN-02 – LoRA Fine-Tuning

Admins may trigger LoRA training jobs.

FR-MTRAIN-03 – Training Status

The app shall display:

- Current status
- Progress
- ETA
- Logs
- Adapter versions

FR-MTRAIN-04 – Teacher Sample Capture

Cloud-generated responses may be stored for future fine-tuning.

FR-MTRAIN-05 – Adapter Registry

The system shall maintain versioned adapter histories.

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15. Admin Features

FR-MADMIN-01 – Settings Management

Admin users shall manage settings from mobile devices.

FR-MADMIN-02 – Prompt Management

Admins may manage prompt templates.

FR-MADMIN-03 – User Management

Admins may:

- Add users
- Remove users
- Change roles
- Lock accounts

FR-MADMIN-04 – Integration Management

Admins may configure:

- LDAP
- Email
- Gemini
- Open-source providers
- Webhooks
- External APIs

FR-MADMIN-05 – Audit Logs

Admins shall access settings audit logs.

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16. Offline & Synchronisation Features

FR-MOFF-01 – Offline Awareness

The app shall detect backend connectivity status.

FR-MOFF-02 – Local Cache Access

Users shall access cached documents and chats while offline.

FR-MOFF-03 – Deferred Synchronisation

Changes made offline shall synchronise automatically.

FR-MOFF-04 – Conflict Resolution

The system shall resolve sync conflicts using configurable policies.

FR-MOFF-05 – Offline AI Mode

Where supported, lightweight local models may answer basic queries offline.

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17. Notifications

FR-MNOTIF-01 – Push Notification Support

The mobile app shall support push notifications.

FR-MNOTIF-02 – Notification Categories

Notifications shall include:

- Upload status
- Chat responses
- Analysis completion
- Repository updates
- Training completion
- Admin alerts

FR-MNOTIF-03 – Notification Preferences

Users may configure notification preferences.

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18. Security Requirements

FR-MSEC-01 – HTTPS Enforcement

All production communication shall use HTTPS.

FR-MSEC-02 – Encrypted Storage

Sensitive mobile data shall be encrypted at rest.

FR-MSEC-03 – Certificate Pinning

The mobile application may support SSL certificate pinning.

FR-MSEC-04 – Offline Security Policy

When offline_only=true no data shall be transmitted externally.

FR-MSEC-05 – Device Integrity Checks

The app may detect rooted or jailbroken devices.

FR-MSEC-06 – Audit Logging

Security-sensitive events shall be logged.

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19. Performance Requirements

FR-MPERF-01 – App Startup

The mobile app shall launch within 5 seconds on supported hardware.

FR-MPERF-02 – Chat Responsiveness

AI chat streaming shall begin within 3 seconds under normal conditions.

FR-MPERF-03 – Background Uploads

Uploads shall continue in the background where OS permissions allow.

FR-MPERF-04 – Battery Optimisation

The application shall minimise battery consumption during background tasks.

FR-MPERF-05 – Efficient Synchronisation

Only changed records shall synchronise where possible.

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20. Mobile API Requirements

FR-MAPI-01 – Shared API Layer

The mobile application shall use the same API ecosystem as the web application.

FR-MAPI-02 – Mobile Optimised Responses

APIs may provide compact mobile response payloads.

FR-MAPI-03 – Pagination

Large datasets shall support pagination.

FR-MAPI-04 – Streaming APIs

Chat and ingestion APIs shall support streaming.

FR-MAPI-05 – API Versioning

The backend shall support versioned APIs.

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21. Data Model Requirements

FR-MDATA-01 – Shared Core Entities

The mobile platform shall use the same logical entities as the web platform including:

- User
- Repository
- Document
- Chunk
- Embedding
- Session
- Message
- Output
- Settings
- TrainingJob

FR-MDATA-02 – Mobile Cache Database

The app shall maintain a local encrypted cache database.

FR-MDATA-03 – Sync Metadata

Local records shall maintain synchronisation metadata.

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22. Accessibility Requirements

FR-MACC-01 – Accessibility Compliance

The mobile app shall support accessibility best practices.

FR-MACC-02 – Screen Reader Support

The app shall support iOS VoiceOver and Android TalkBack.

FR-MACC-03 – Dynamic Font Scaling

Text shall support device font scaling.

FR-MACC-04 – High Contrast Support

The app shall support high contrast modes.

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23. Deployment Requirements

FR-MDEP-01 – Android Distribution

The Android app shall support:

- APK distribution
- Managed enterprise deployment
- Google Play deployment

FR-MDEP-02 – iOS Distribution

The iOS app shall support:

- Apple App Store
- Enterprise deployment
- TestFlight

FR-MDEP-03 – Configuration Profiles

Environment configurations shall support:

- Development
- Test
- Staging
- Production
- Offline-secure mode

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24. Future Enhancements

The architecture shall support future enhancements including:

- On-device LLM inference
- Federated AI learning
- Real-time collaboration editing
- AR document overlays
- Video analysis
- IoT integration
- GIS integration
- Enterprise workflow automation
- Smart meeting transcription
- AI agents

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25. Conclusion

The DocTel Mobile Application shall provide a secure, scalable, enterprise-grade mobile AI platform delivering full feature parity with the DocTel web platform while extending support for multiple open-source AI technologies.

The solution shall support privacy-first deployments, offline-aware workflows, enterprise governance, and flexible AI model orchestration suitable for ZETDC operational requirements.

This specification defines the baseline functional requirements for implementation, testing, deployment, and future enhancement of the DocTel Mobile platform.

End of Document